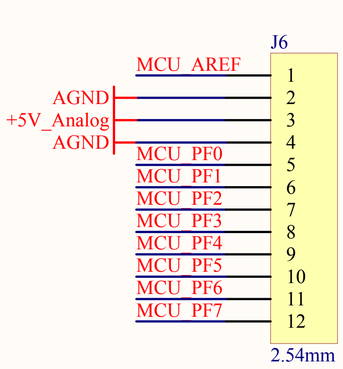
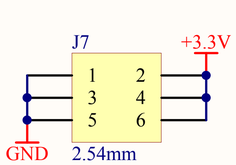
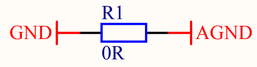
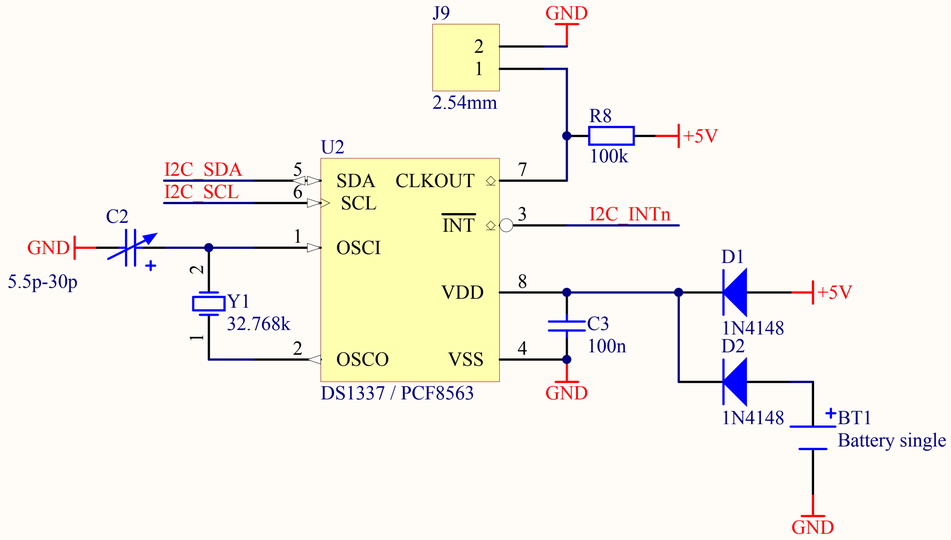


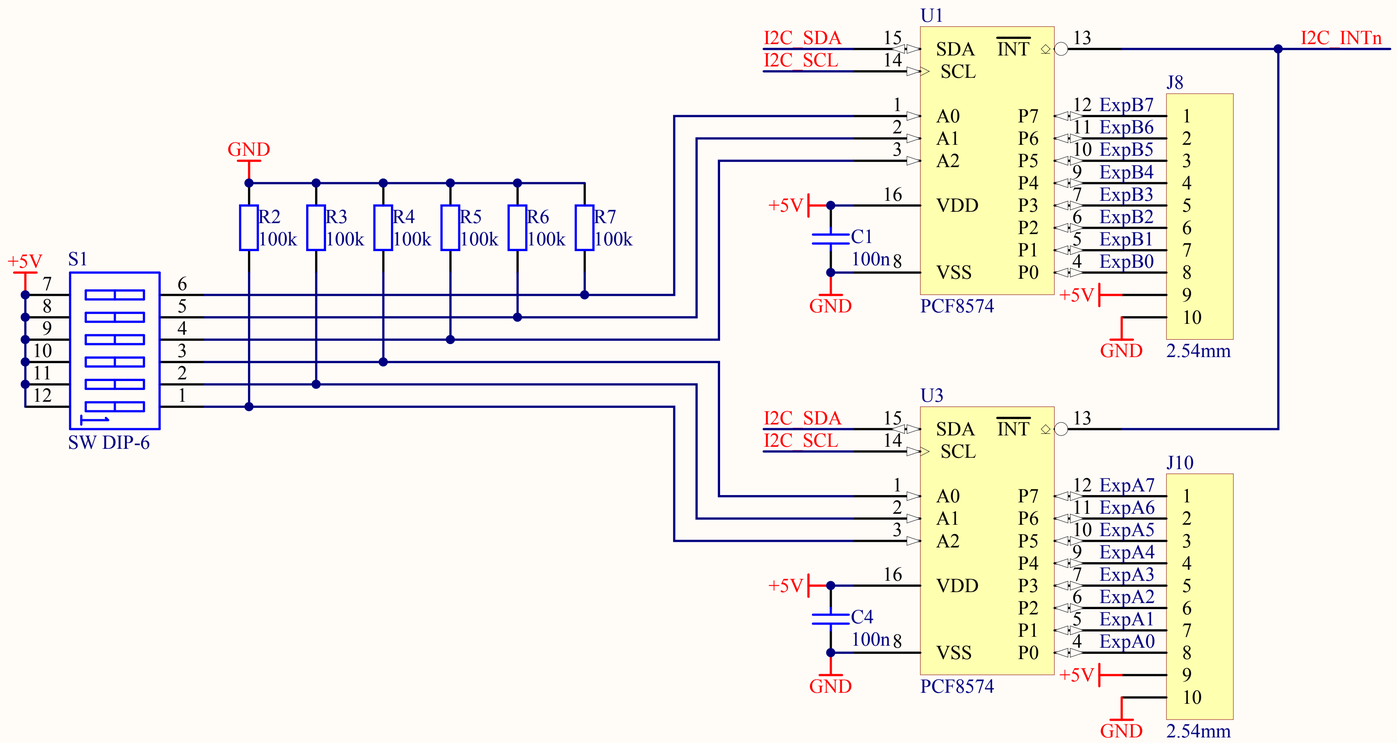


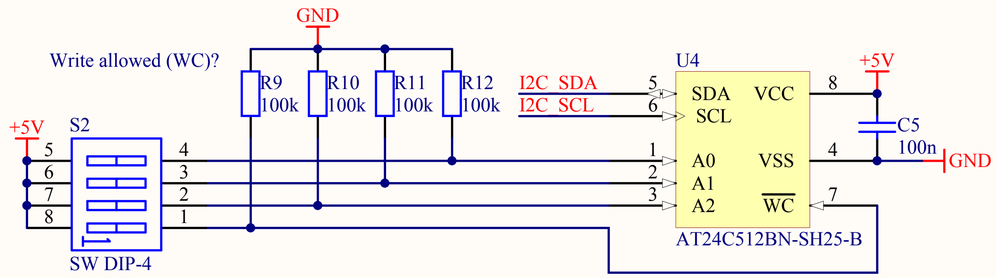




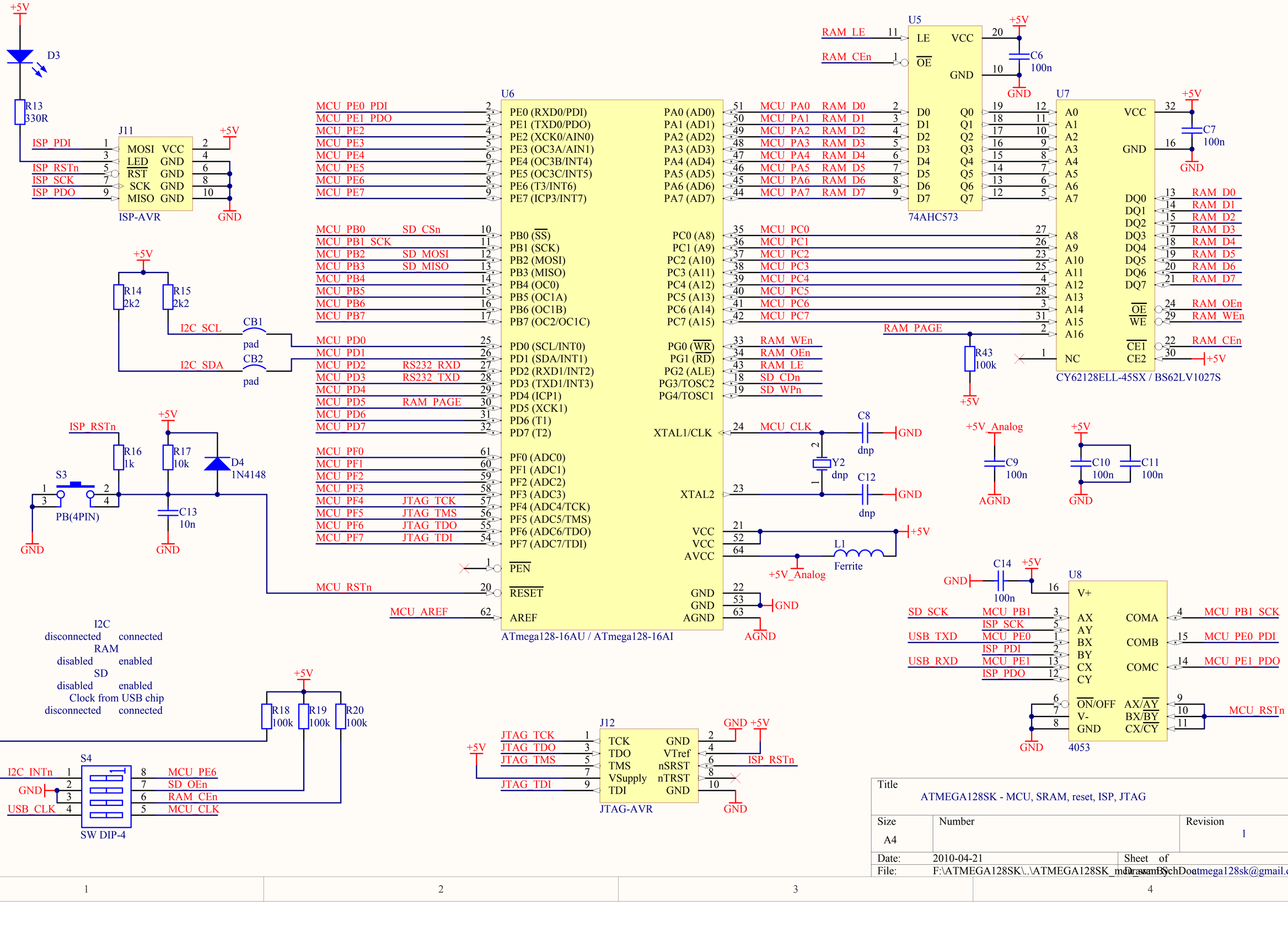
Title: ATMEGA128SK - GPIO		
Size: A4	Number:	Revision: 1
Date: 2010-04-21		Sheet of
File: F:\ATMEGA128SK\..\ATMEGA128SK_gpio.SchDoc	Drawn By:	atmega128sk@gmail.com

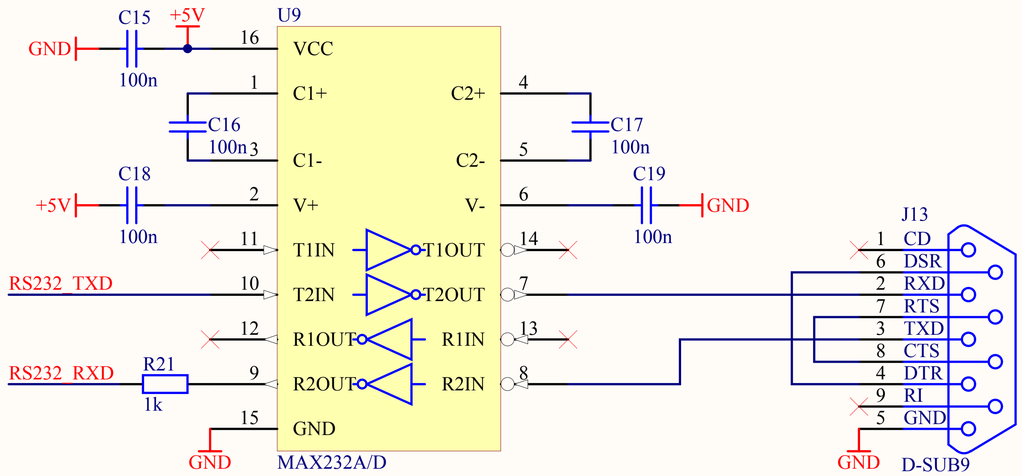




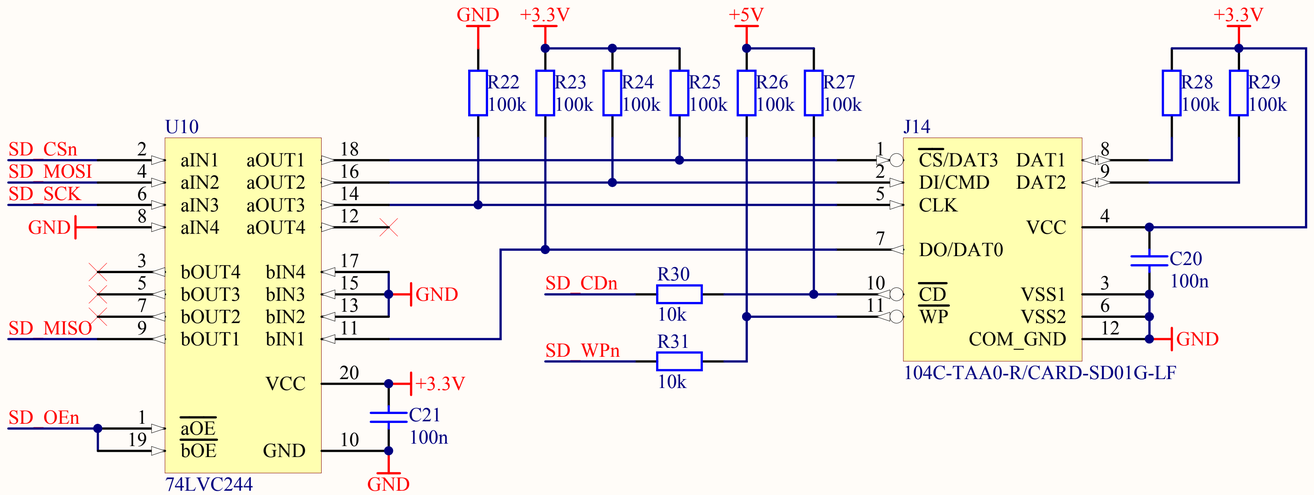


Title: ATMEGA128SK - serial EEPROM, expanders, RTC		
Size: A4	Number:	Revision: 1
Date: 2010-04-21	Sheet of	
File: F:\ATMEGA128SK\..\ATMEGA128SK_i2c.SchDoc	Drawn by:	atmega128sk@gmail.com

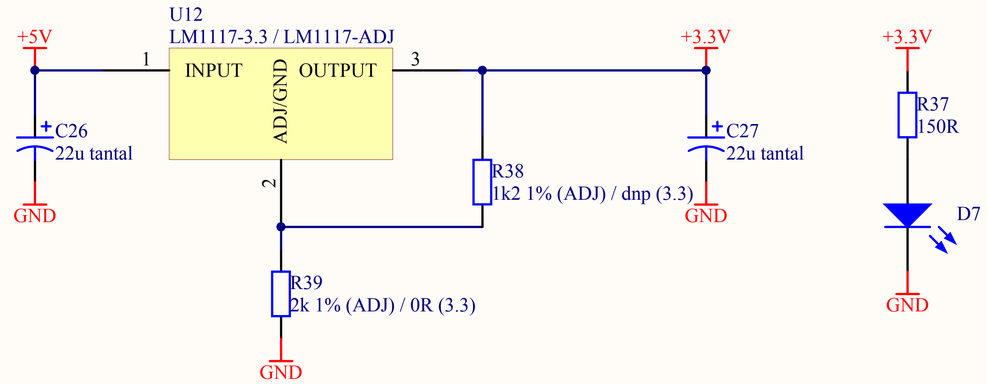

+5V
D3
R13
330R
J11
ISP_PDI
ISP_RSTn
ISP_SCK
ISP_PDO
MOSI VCC
LED GND
RST GND
SCK GND
MISO GND
ISP-AVR
GND
U6
MCU_PE0_PDI
MCU_PE1_PDO
MCU_PE2
MCU_PE3
MCU_PE4
MCU_PE5
MCU_PE6
MCU_PE7
PE0 (RXD0/PDI)
PE1 (TXD0/PDO)
PE2 (XCK0/AIN0)
PE3 (OC3A/AIN1)
PE4 (OC3B/INT4)
PE5 (OC3C/INT5)
PE6 (T3/INT6)
PE7 (ICP3/INT7)
MCU_PB0 SD_CSn
MCU_PB1_SCK
MCU_PB2 SD_MOSI
MCU_PB3 SD_MISO
MCU_PB4
MCU_PB5
MCU_PB6
MCU_PB7
PB0 (SS)
PB1 (SCK)
PB2 (MOSI)
PB3 (MISO)
PB4 (OC0)
PB5 (OC1A)
PB6 (OC1B)
PB7 (OC2/OC1C)
MCU_PD0
MCU_PD1
MCU_PD2 RS232_RXD
MCU_PD3 RS232_TXD
MCU_PD4
MCU_PD5 RAM_PAGE
MCU_PD6
MCU_PD7
PD0 (SCL/INT0)
PD1 (SDA/INT1)
PD2 (RXD1/INT2)
PD3 (TXD1/INT3)
PD4 (ICP1)
PD5 (XCK1)
PD6 (T1)
PD7 (T2)
MCU_PF0
MCU_PF1
MCU_PF2
MCU_PF3
MCU_PF4 JTAG_TCK
MCU_PF5 JTAG_TMS
MCU_PF6 JTAG_TDO
MCU_PF7 JTAG_TDI
PF0 (ADC0)
PF1 (ADC1)
PF2 (ADC2)
PF3 (ADC3)
PF4 (ADC4/TCK)
PF5 (ADC5/TMS)
PF6 (ADC6/TDO)
PF7 (ADC7/TDI)
PEN
RESET
MCU_RSTn
AREF
MCU_AREF
PA0 (AD0)
PA1 (AD1)
PA2 (AD2)
PA3 (AD3)
PA4 (AD4)
PA5 (AD5)
PA6 (AD6)
PA7 (AD7)
PC0 (A8)
PC1 (A9)
PC2 (A10)
PC3 (A11)
PC4 (A12)
PC5 (A13)
PC6 (A14)
PC7 (A15)
PG0 (WR)
PG1 (RD)
PG2 (ALE)
PG3/TOSC2
PG4/TOSC1
XTAL1/CLK
XTAL2
VCC
VCC
AVCC
GND
GND
AGND
ATmega128-16AU / ATmega128-16AI
MCU_PA0 RAM_D0
MCU_PA1 RAM_D1
MCU_PA2 RAM_D2
MCU_PA3 RAM_D3
MCU_PA4 RAM_D4
MCU_PA5 RAM_D5
MCU_PA6 RAM_D6
MCU_PA7 RAM_D7
MCU_PC0
MCU_PC1
MCU_PC2
MCU_PC3
MCU_PC4
MCU_PC5
MCU_PC6
MCU_PC7
RAM_WEn
RAM_OEn
RAM_LE
SD_CDn
SD_WPn
MCU_CLK
U5
RAM_LE
RAM_CEn
LE VCC
OE
GND
74AHC573
C6
100n
U7
RAM_PAGE
R43
100k
A0 VCC
A1
A2
GND
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
NC
DQ0
DQ1
DQ2
DQ3
DQ4
DQ5
DQ6
DQ7
OE
WE
CE1
CE2
RAM_D0
RAM_D1
RAM_D2
RAM_D3
RAM_D4
RAM_D5
RAM_D6
RAM_D7
RAM_OEn
RAM_WEn
RAM_CEn
CY62128ELL-45SX / BS62LV1027S
C7
100n
R14
2k2
R15
2k2
I2C_SCL
I2C_SDA
CB1
CB2
pad
C8
dnp
Y2
dnp
C12
dnp
+5V_Analog
C9
100n
AGND
C10
100n
C11
100n
L1
Ferrite
ISP_RSTn
R16
1k
R17
10k
D4
1N4148
S3
PB(4PIN)
C13
10n
C14
100n
U8
V+
AX
AY
BX
BY
CX
CY
ON/OFF
V-
GND
COMA
COMB
COMC
AX/AY
BX/BY
CX/CY
4053
SD_SCK
USB_TXD
USB_RXD
MCU_PB1_SCK
MCU_PE0_PDI
MCU_PE1_PDO
MCU_RSTn
I2C
disconnected connected
RAM
disabled enabled
SD
disabled enabled
Clock from USB chip
disconnected connected
R18
100k
R19
100k
R20
100k
S4
I2C_INTn
USB_CLK
MCU_PE6
SD_OEn
RAM_CEn
MCU_CLK
SW DIP-4
J12
JTAG_TCK
JTAG_TDO
JTAG_TMS
JTAG_TDI
TCK GND
TDO VTref
TMS nSRST
VSupply nTRST
TDI GND
JTAG-AVR
ISP_RSTn
Title
ATMEGA128SK - MCU, SRAM, reset, ISP, JTAG
Size
A4
Number
Revision
1
Date: 2010-04-21
Sheet of
File: F:\ATMEGA128SK\..\ATMEGA128SK_mdrasm.SchDoc
Drawn By: atmega128sk@gmail.com

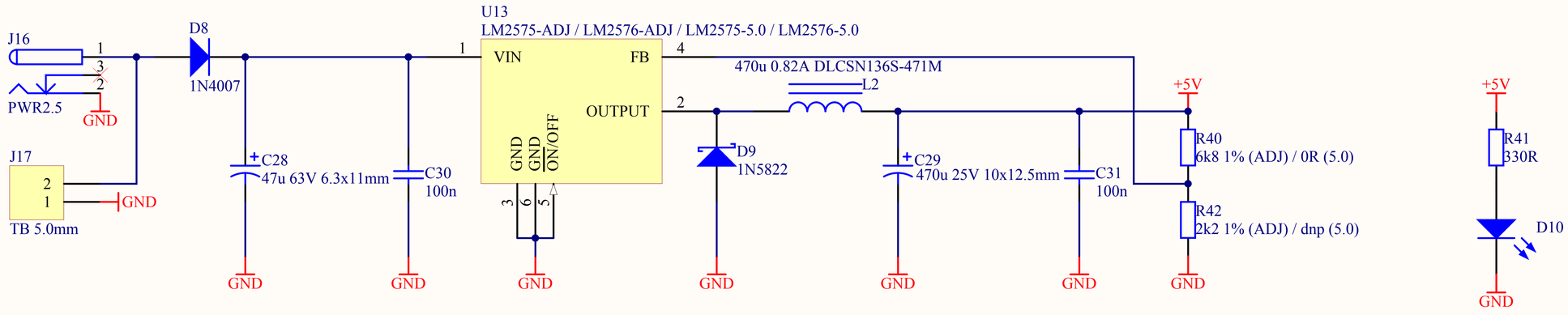


Title: ATMEGA128SK - RS232 interface		
Size: A4	Number:	Revision: 1
Date: 2010-04-21	Sheet of	
File: F:\ATMEGA128SK\..\ATMEGA128SK_rs232.SchDoc	Drawn By: atmega128sk@gmail.com	



Title	ATMEGA128SK - SD card	
Size: A4	Number:	Revision: 1
Date: 2010-04-21	Sheet of	
File: F:\ATMEGA128SK\..\ATMEGA128SK_sdDoc.SchDoc	Drawn By:	atmega128sk@gmail.com

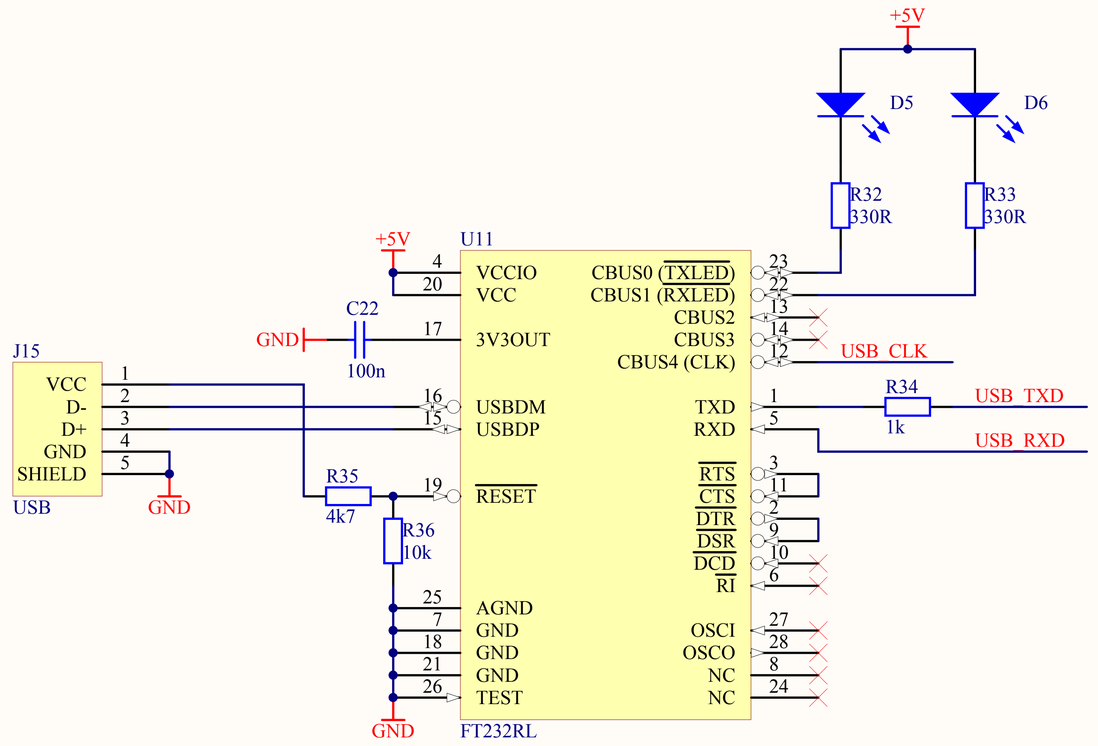




Switchers made simple:

Iout_max = 0.7A
Vin_min = 7.0 V
Vin_max = 30V
L = 470 uH, 0.8A
Cin = 47 uF (43V)
Cout = 330 uF (20V, 0.15 Ohm > ESR > 0.075 Ohm)
ID = 3A (Schottky) 1N5822, 31DQ04, MBR340P

Title: ATMEGA128SK - power supply		
Size: A4	Number:	Revision: 1
Date: 2010-04-21		Sheet of
File: F:\ATMEGA128SK\..\ATMEGA128SK_supply.SchDoc	Drawn By:	atmega128sk@gmail.com



Title	ATMEGA128SK - USB interface	
Size: A4	Number:	Revision: 1
Date: 2010-04-21	Sheet of	
File: F:\ATMEGA128SK\..\ATMEGA128SK_usb.sch	Drawn By:	atmega128sk@gmail.com